Prowler’s State of Cloud Security Report 2025

Toni de la Fuente June 4, 2025

We’ve been in this industry long enough to know that survey data rarely tells the whole story. However, when Prowler surveyed 655 security professionals, some truths emerged that we need to discuss.

Here's what matters: 96% of security teams say they're confident in their cloud security.

That's great!

But let's focus on the 4% who aren't—they're telling us something important. They're worried about shadow IT, inadequate tooling, and spiraling costs. These aren't edge cases; they're leading indicators of industry-wide challenges.

Key Takeaways

The State of Cloud Security 2025 report reveals an industry at an inflection point. While confidence remains high and new technologies drive significant improvements, fundamental challenges persist. Organizations must balance innovation with security, cost optimization with comprehensive protection, and automation with human expertise.

Success in 2025 and beyond will require:

1. **Embracing open cloud security tools** to drive innovation and collaboration
2. **Investing in automation** to save time and improve efficiency
3. **Addressing the talent gap** through training and strategic hiring
4. **Optimizing costs** without compromising security
5. **Leveraging AI and ML** for enhanced threat detection and response

As cloud environments become more complex and threats more sophisticated, the organizations that thrive will be those that can adapt quickly, invest wisely, and maintain a proactive approach to security. The tools and technologies are available – the challenge is implementing them effectively while managing costs and compliance requirements.

The Multi-Cloud Reality Check

- 64% hybrid-cloud
- 55% multi-cloud
- Only 17% single-cloud

This isn't just about technical complexity. It's about organizational maturity. We've moved from "should we go to the cloud?" to "how do we manage security across multiple clouds?" That's progress, but it demands new thinking.

Open Source is Winning

88% are using open cloud security tools. Why? Because closed systems can't keep pace with modern threats. When 86% report measurable security improvements from open tools,

that's not just adoption – that's validation.

But here's the kicker: It's not just about better security. It's about better collaboration (83% report this) and cost optimization (80% see savings). Open source isn't just technically superior; it's organizationally transformative.

The Automation Imperative

Companies save 19 hours per week with automation. That's not just efficiency – the difference between playing defense and offense. Are 25% still doing manual processes? They're not just behind; they're at risk.

Hard Truths About Compliance

37% failed compliance audits last year. Let that sink in. We have all the frameworks (NIST, ISO 27001, SOC 2), but implementation is failing. This isn't a tool problem; it's an execution problem.

The Path Forward

Stop treating security as a cost center. The companies succeeding are investing 27% more in compliance this year. They're using AI (79% adoption), embracing open source, and automating everything they can.

The future belongs to those who can balance speed with security, openness with control, and automation with human judgment. Everything else is just noise.

The findings paint a picture of an industry simultaneously confident and concerned, embracing cutting-edge technologies while grappling with fundamental challenges. Let's dive into the key insights shaping cloud security in 2025.

Cloud Environments: Hybrid and Multi-Cloud Dominate

Perhaps unsurprisingly, cloud environments are now the default, with 64% of organizations operating in hybrid-cloud environments and 55% in multi-cloud setups. The days of simple, single-cloud deployments are increasingly rare, with only 17% of organizations still operating in a single-cloud environment.

This shift toward complexity brings both opportunities and challenges. While organizations benefit from increased flexibility and redundancy, they must also manage security across

multiple platforms and environments, requiring sophisticated tools and expertise, which is where Cloud Security Posture Management (CSPM) solutions come in.

Confidence is High, But Not Universal

An impressive 96% of security professionals feel fully confident in their company's ability to manage cloud security end-to-end. This statistic initially appears reassuring, suggesting that organizations have largely mastered cloud security fundamentals. However, dig deeper, and a more nuanced picture emerges.

Among the 4% who aren't fully confident, the concerns are telling:

- 79% worry about unauthorized cloud services bypassing security measures
- 71% don't believe their current tools can cover all aspects of their cloud environment
- 68% cite the cost of operationalizing cloud security tools as too high

These concerns highlight that even as overall confidence remains high, significant challenges persist beneath the surface.

Open Cloud Security Tools: Driving Innovation and Collaboration

One of the most striking findings is the widespread adoption of open cloud security tools. A remarkable 88% of organizations now use open cloud security tools, and the benefits are clear:

- 9 in 10 users agree that these tools are a primary driver of innovation in their security programs
- 83% report strengthened collaboration between security and IT departments
- 8 in 10 users have seen reduced cloud operating costs through better identification of redundancies

Perhaps most importantly, 86% have seen a measurable reduction in security breaches due to better visibility and proactive security measures. These statistics underscore how open-source solutions are not just cost-effective alternatives but are actively driving better security outcomes.

Automation: The Time-Saving Game Changer

Automation has become a critical component of cloud security strategies. While 44% of organizations say their process for monitoring and managing cloud security is highly automated, and 31% claim full automation, the benefits are substantial.

Organizations using fully or highly automated processes save an average of 19 hours per week – nearly half a work week. This time savings allows security teams to focus on strategic initiatives rather than manual, repetitive tasks.

Interestingly, the 25% of organizations still using manual processes are actively looking to change, with 85% planning to automate within the next 12 months. The manufacturing, transport, and logistics industries are particularly behind in automation adoption, being 88% more likely than average to use manual processes still.

The Vulnerability Challenge

Despite regular assessments, vulnerability management remains a significant challenge. While 96% of organizations conduct regular vulnerability assessments, and 67% do so monthly or more frequently, 46% still struggle with vulnerability management.

The scope of the challenge is evident: organizations detect an average of 17 vulnerabilities in their cloud environments per week. The top challenges in managing these vulnerabilities include:

- Budget constraints (35%)
- Integrating vulnerability management with existing workflows (34%)
- Lack of skilled personnel (32%)

Compliance: A Mixed Picture

Compliance remains a critical concern, with 78% of organizations employing governance frameworks for cloud security. The most popular frameworks include:

- NIST Cybersecurity Framework (40%)
- ISO/IEC 27001 (31%)
- SOC 2 Compliance (28%)

However, 44% of organizations struggle to comply with new cloud security regulations, and 37% have failed a compliance audit due to cloud security issues in the past 12 months. This

disconnect between framework adoption and actual compliance success suggests that having the right frameworks isn't enough – effective implementation and ongoing management are crucial.

Organizations are investing more in compliance to address these challenges, with 75% planning to spend more on compliance-related measures in 2025 than in 2024, averaging a 27% increase.

The AI Revolution in Cloud Security

Artificial Intelligence is transforming cloud security. 79% of organizations use AI technologies to monitor and manage cloud security, reaping benefits such as:

- Human augmentation (44%)
- Improved data protection (42%)
- Strengthened threat detection and response capabilities (38%)

Looking Ahead: The Future of Cloud Security

As we look toward the future, several trends are emerging. Security professionals anticipate key resource gaps in the next 12 months, including limited budgets (45%), talent shortages (42%), and insufficient automation (34%).

To address cost concerns, 86% of organizations are taking steps to optimize costs in their cloud security strategy, with strategies including:

- Automating security processes to reduce manual effort (60%)
- Consolidating security tools and platforms (51%)
- Scaling security measures based on risk assessments (43%)

When asked about technologies that will have the greatest impact on cloud security in the next three years, respondents highlighted:

- AI and ML-driven security analytics (27%)
- Open source cloud security tools and technologies (17%)
- Automated threat intelligence and response (16%)

Getting Started with Prowler for CSPM

Managing security doesn't have to be a nightmare. With Prowler, you get a powerful, open-source, and automated CSPM solution that simplifies cloud security and compliance.

So why wait?

Start using Prowler today and take control of your security with confidence.

Try Prowler Cloud Free.